How large is the missing rho price?

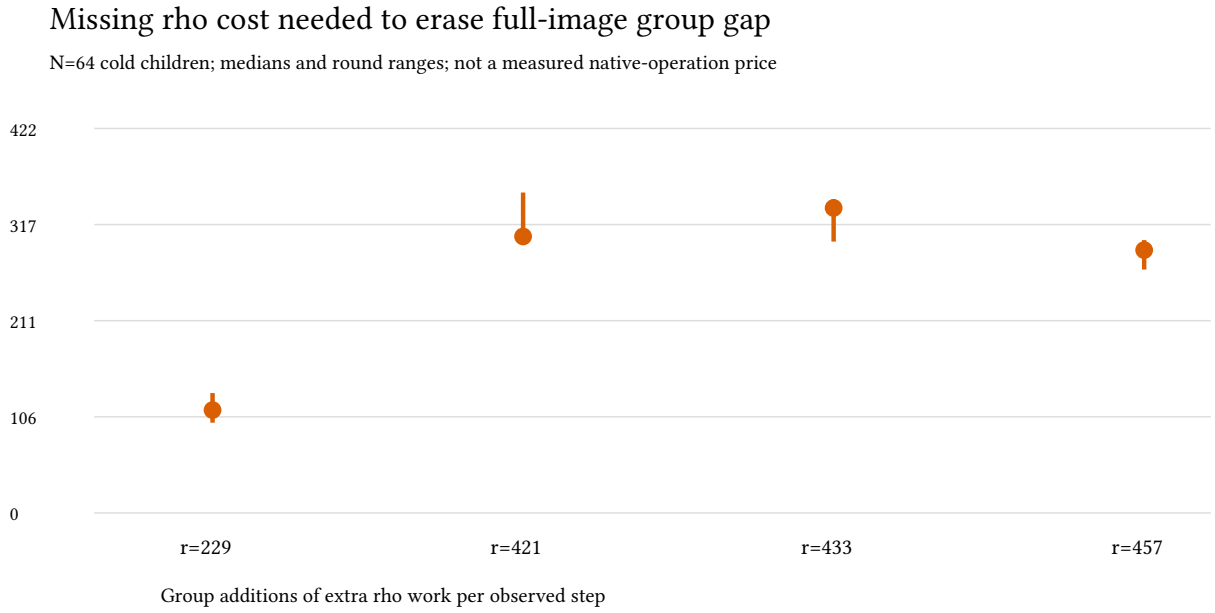
Frozen native derivation, 2026-10-15. Four complete-solve count files, their independent replay certificates and 48 paired prefix/round rows passed hash, key, scalar-count and group-total checks. No solver input or child was changed. Host: Darwin arm64 shared L0; elapsed time is not used as a performance result.

For each child, let G_{full} and G_{rho} count cold group adds and doubles from validation through final verification, and let R_{steps} count all observed rho walk steps including retries.

$W_{\text{break}} = \frac{G_{\text{full}} - G_{\text{rho}}}{R_{\text{steps}}}$.

This is the **extra omitted rho charge per observed step** needed to tie the full-image group-count lower bound if every omitted full-image operation is priced at zero. A positive omitted full-image price makes the required rho charge larger. It is not a measured native conversion, complete portable S, or end-to-end speed ratio.

Order r	N64 median GAE/step	Round min	Round max
229	112.973	99.105	131.619
421	303.573	302.425	351.781
433	334.844	297.947	344.670
457	288.595	267.223	299.550



Across every prefix and round, the minimum is 99.105 GAE/step at N64, order 229; the maximum is 46,594.667 at N1, order 421. The N1 threshold charges the cold full-image census to one solve.

The source count data, exact run hashes and replay certificates are in `../TREND_V1.json`, `PROTOCOL_V1.md` and `RUN.md`; all 48 calculated rows are in `SENSITIVITY_V1.json`. The native source is `examples/different_j_price_sensitivity.rs`. The parent report retains the matched BSGS and rho rows, separate Fp multiplication/inversion axes and memory. Complete portable S still requires disjoint native operation counts, calibration across host classes and independent trace coverage; isolated wall and second-host performance are separate gates. A direct chi-coordinate oracle that avoids the source census is the structural next test.